Instructor's Note:

`bitwhiz` is the reference hydrocode for ME 697, Intermediate Shock and Detonation Theory (Purdue, Fall 2026): a one-dimensional, planar, Lagrangian code for plate impact, with shock and release waves crossing layers of solids and liquids, and spall where release waves meet in tension. Two solution methods share one code: this sheet is Method RS (a Riemann solver), and the companion sheet is Method AV (artificial viscosity). It was written by D. Kittell with the assistance of Claude Opus 5.5 (Anthropic), an AI model; the instructor directed, reviewed, and tested every part of it and is responsible for its content.

These equations build upon U3. Continuum Mechanics & Balance Laws, U7. The Riemann Problem for Compressible Fluids (Acoustic Solution), and the Method AV sheet (units, equation of state, spall)

Notation & Scope:

- One-dimensional hydrocode, planar, Lagrangian, and purely hydrodynamic, such that $\boldsymbol{\sigma_{xx} = -P}$ with $\boldsymbol{Y = 0}$ and ***no artificial viscosity***
- Cell-Centered Grid, $\boldsymbol{\rho}$, $\mathrm{V} = \boldsymbol{1/\rho}$, $\boldsymbol{u}$, $\boldsymbol{e}$, $\boldsymbol{E = e + \frac{1}{2}u^2}$, $\boldsymbol{P}$, $\boldsymbol{c}$, $\boldsymbol{Z = \rho c}$ live in the cells, ***velocity included***; a face carries its position $\boldsymbol{x}$ and the Riemann solution $\boldsymbol{u_\star}$, $\boldsymbol{P_\star}$
- Indices, cells $\boldsymbol{k}$, faces $\boldsymbol{k \pm \frac{1}{2}}$; the cell state is at whole steps $\boldsymbol{n}$; $\boldsymbol{u_\star}$ and $\boldsymbol{P_\star}$ hold for the whole step $\boldsymbol{n \to n+1}$, so they carry $\boldsymbol{n + \frac{1}{2}}$
- Face States, $\boldsymbol{u_k^L}$, $\boldsymbol{P_k^L}$ and $\boldsymbol{u_k^R}$, $\boldsymbol{P_k^R}$: what cell $\boldsymbol{k}$ offers the Riemann problems on its left ($\boldsymbol{k - \frac{1}{2}}$) and right ($\boldsymbol{k + \frac{1}{2}}$) faces (Step 1)
- Cell Mass, $\boldsymbol{M_k}$ per unit area, fixed; the areal mass coordinate $\boldsymbol{h}$ of U7 advances by $\boldsymbol{M_k}$ per cell, and faces are lines $\boldsymbol{h} =$ const.
- Star Region, $\boldsymbol{u_\star}$ and $\boldsymbol{P_\star}$ are continuous across the contact (U7); $\mathrm{V}_{\boldsymbol{\star L}}$ and $\mathrm{V}_{\boldsymbol{\star R}}$ are not, and this method never needs them
- Boxes titled **bitwhiz:** go beyond the ***project rules***, to what the reference code `bitwhiz` does (deck: `method = RS`)

Units and Shared Pieces (from the Method AV sheet):

- Units, $\boldsymbol{x}$ [mm], $\boldsymbol{t}$ [$\boldsymbol{\mu}$s], $\boldsymbol{\rho}$ [g/cm$^{\mathbf{3}}$], $\boldsymbol{u}$, $\boldsymbol{c}$ [km/s], $\boldsymbol{P}$ [GPa], $\boldsymbol{e}$ [MJ/kg], $\boldsymbol{M_k}$ [g/cm$^{\mathbf{3}}\cdot$mm], $\boldsymbol{Z}$ [g/cm$^{\mathbf{3}}\cdot$km/s] so $\boldsymbol{Z\,u}$ [GPa]
- Shared, the mesh of Step 0, the equation of state $\boldsymbol{P\,(\rho, e)}$ and sound speed $\boldsymbol{c}$, the void boundary, and (`bitwhiz`) the spall rule and expansion branch: ***the same as Method AV***

Godunov's Method, and Its Relatives (a Map, Not a Recipe):

- Godunov (1959), keep one ***average*** state per cell; at every face, solve the Riemann problem between the two neighbor cells; use its solution as the ***flux*** through that face for one step; update each cell from its two faces. Conservative, monotone, first order; shocks are captured without $\boldsymbol{q}$, smeared over a few cells
- The Riemann Solver, ***what goes at the face***: exact (iterative, U7); **acoustic** (this sheet: $\boldsymbol{Z}$ frozen for one step, no iteration); HLL/HLLC (estimated wave speeds; in $\boldsymbol{h}$ with speeds $\boldsymbol{-Z_L}$, $\boldsymbol{+Z_R}$, the HLLC star state ***is*** the acoustic one); two-shock (Dukowicz 1985: $\boldsymbol{Z \approx \rho\,(c + s_1\,|u_\star - u|)}$, a shock impedance from $\boldsymbol{U_S}$–$\boldsymbol{u_p}$ data)
- Reconstruction, ***what states the face sees***: first order uses the cell averages; MUSCL (van Leer 1979, first written for Lagrangian gas dynamics) puts a limited linear slope in each cell, second order; WENO5 blends three quadratics, fifth order where smooth. Both sharpen shocks; in general both need two ghost cells per end (`bitwhiz`'s order 2 is MUSCL-Hancock with Van Leer's limiter, Step 1)
- Time Integration, first order is forward Euler with the face fluxes held over the step; with MUSCL, use a MUSCL-Hancock half step or a strong-stability-preserving Runge-Kutta (RK2/RK3) so the time order matches the space order
- Splitting, Strang splitting combines separate operators (hydro, then a reaction or a second direction) to second order; one-dimensional inert hydro needs none
- References, Godunov, *Mat. Sb.* **47** (1959); van Leer, *J. Comput. Phys.* **32** (1979); Dukowicz, *J. Comput. Phys.* **61** (1985); Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics* (Springer, 2009); Kittell et al., *Propellants Explos. Pyrotech.* **45** (2020) 295–315

Step 0. Mesh and Initial Conditions:

- Faces and Cell Mass, as in Method AV: ***every material interface lands on a cell face***, $M_k = \rho_{0,k}\left(x^0_{k+\frac{1}{2}} - x^0_{k-\frac{1}{2}}\right)$
- Cell State, $\rho^0_k = \rho_{0,k}$, $e^0_k = 0$, $P^0_k = 0$, $c^0_k = c_{0,k}$, $u^0_k = u_{0,k}$ the cell's own initial velocity
- Impact Face, nothing special: two cells with different velocities meet at a face, which ***is*** a Riemann problem; no momentum-weighted starting value is needed

Step 1. Face States, What Each Cell Offers Its Faces (`order` = 1 or 2):

- Order 1, (the project's Option B) $u^L_k = u^R_k = u^n_k$, $P^L_k = P^R_k = P^n_k$ the cell averages: first-order Godunov
- Order 2 (`bitwhiz`, the default), MUSCL-Hancock. For $W = u$ and $W = P$, the slopes per unit mass to each neighbor

  . $a = \dfrac{W_k - W_{k-1}}{\frac{1}{2}(M_{k-1} + M_k)}$ $b = \dfrac{W_{k+1} - W_k}{\frac{1}{2}(M_k + M_{k+1})}$ all at n

- Van Leer Limiter, $s_W = \dfrac{2ab}{a+b}$ if $ab > 0$, else $s_W = 0$ a peak or trough gets no slope, so no new extremum (no ringing)
- Half Step, each face value advanced by $\frac{1}{2}\Delta t^n$ with the cell's own acoustic equations, $\partial u/\partial t = -\partial P/\partial h$, $\partial P/\partial t = -Z^2\,\partial u/\partial h$

  . $u^{L,R}_k = u^n_k \mp \dfrac{M_k}{2} s_u - \dfrac{\Delta t^n}{2} s_P$

  . $P^{L,R}_k = P^n_k \mp \dfrac{M_k}{2} s_P - \dfrac{\Delta t^n}{2} (Z^n_k)^2 s_u$

- Only u and P, the acoustic solver takes Z from the cell averages, and u, P are continuous across a material interface, so a slope may span one; V, which jumps there, is never reconstructed
- Ends, a cell beside a ghost gets $s_u = s_P = 0$: the free surfaces stay first order, and one ghost cell per end is enough

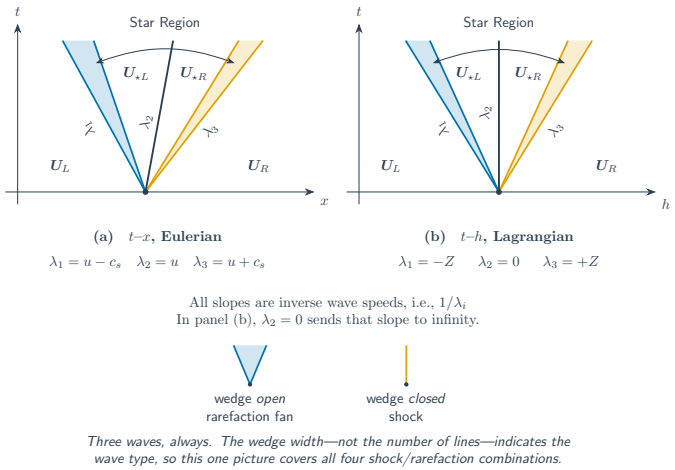


Figure 1: The Riemann problem at one face (from U7; representational only). In the Lagrangian frame (b) the face is the line $\boldsymbol{h} =$ const.; the left and right waves leave it at $-\boldsymbol{Z_L}$ and $+\boldsymbol{Z_R}$, and the face itself carries $\boldsymbol{u_\star}$ and $\boldsymbol{P_\star}$.

Step 2. The Riemann Problem at Every Face (Acoustic Solution, U7):

- Left and Right, for the face $\boldsymbol{k+\frac{1}{2}}$: $\boldsymbol{u_L = u_k^R}$, $\boldsymbol{P_L = P_k^R}$ from cell $\boldsymbol{k}$, $\boldsymbol{u_R = u_{k+1}^L}$, $\boldsymbol{P_R = P_{k+1}^L}$ from cell $\boldsymbol{k+1}$ (Step 1); impedances $\boldsymbol{Z_L = \rho_k^n c_k^n}$, $\boldsymbol{Z_R = \rho_{k+1}^n c_{k+1}^n}$, held fixed for the step

- Riemann Invariants, $\boldsymbol{\Phi_1 = P_L + Z_L u_L}$ $\boldsymbol{\Phi_2 = P_R - Z_R u_R}$

- Star State, $\boldsymbol{u_{\star,k+\frac{1}{2}}^{n+\frac{1}{2}} = \dfrac{\Phi_1 - \Phi_2}{Z_L + Z_R}}$ $\boldsymbol{P_{\star,k+\frac{1}{2}}^{n+\frac{1}{2}} = \dfrac{Z_R\Phi_1 + Z_L\Phi_2}{Z_L + Z_R}}$

- Expanded, $\boldsymbol{u_\star = \dfrac{Z_L u_L + Z_R u_R + P_L - P_R}{Z_L + Z_R}}$

  . $\boldsymbol{P_\star = \dfrac{Z_R P_L + Z_L P_R + Z_L Z_R (u_L - u_R)}{Z_L + Z_R}}$

- Reading $\boldsymbol{P_\star}$, for $\boldsymbol{Z_L = Z_R = Z}$: $\boldsymbol{P_\star = \dfrac{P_L + P_R}{2} + \dfrac{Z}{2}(u_L - u_R)}$ the average, plus a ***built-in viscosity*** $\boldsymbol{\propto \rho c \Delta u}$

- Impact Check, $\boldsymbol{P_L = P_R = 0}$, $\boldsymbol{u_R = 0}$: $\boldsymbol{u_\star = \dfrac{Z_L u_L}{Z_L + Z_R}}$, $\boldsymbol{P_\star = \dfrac{Z_L Z_R u_L}{Z_L + Z_R}}$ the U6 impedance match, with acoustic $\boldsymbol{Z}$

- Note: frozen $\boldsymbol{Z}$ is a weak-wave approximation ***at one face for one step***; the shock speed and jump still come out right, because Steps 3–4 conserve mass, momentum, and energy exactly

Step 3. Positions and Densities:

- Face Motion, $\boldsymbol{x_{k+\frac{1}{2}}^{n+1} = x_{k+\frac{1}{2}}^{n} + u_{\star,k+\frac{1}{2}}^{n+\frac{1}{2}} \Delta t^{n}}$ the face moves with the contact
- Cell Density, $\boldsymbol{\rho_k^{n+1} = \dfrac{M_k}{x_{k+\frac{1}{2}}^{n+1} - x_{k-\frac{1}{2}}^{n+1}}}$ identical to the volume flux update $\mathrm{V}_k^{n+1} = \mathrm{V}_k^n + \dfrac{\Delta t^n}{M_k}\left(u_{\star,k+\frac{1}{2}} - u_{\star,k-\frac{1}{2}}\right)$

Step 4. Velocity, Energy, and Pressure (Each Cell from Its Two Faces):

- Momentum, $\boldsymbol{u_k^{n+1} = u_k^n - \dfrac{\Delta t^n}{M_k}\left(P_{\star,k+\frac{1}{2}} - P_{\star,k-\frac{1}{2}}\right)}$
- Total Energy, $\boldsymbol{E_k^{n+1} = E_k^n - \dfrac{\Delta t^n}{M_k}\left[(P_\star u_\star)_{k+\frac{1}{2}} - (P_\star u_\star)_{k-\frac{1}{2}}\right]}$
- Internal Energy, $\boldsymbol{e_k^{n+1} = E_k^{n+1} - \dfrac{1}{2}\left(u_k^{n+1}\right)^2}$
- Pressure and Sound Speed, $\boldsymbol{P_k^{n+1} = P\left(\rho_k^{n+1}, e_k^{n+1}\right)}$, kept $\boldsymbol{\geq 0}$ (`bitwhiz`: the spall rule), then $\boldsymbol{c_k^{n+1}}$ at the pressure kept as in Method AV
- Conservation, each face flux leaves one cell and enters its neighbor, so the sums telescope: total mass, momentum $\sum \boldsymbol{M_k u_k}$, and energy $\sum \boldsymbol{M_k E_k}$ are conserved ***to round-off*** ($\boldsymbol{P_\star = 0}$ at both free surfaces)

Time Step (CFL Condition):

- Per Cell, $\boldsymbol{\Delta t_k = \dfrac{x_{k+\frac{1}{2}}^{n+1} - x_{k-\frac{1}{2}}^{n+1}}{c_k^{n+1}} = \dfrac{M_k}{Z_k^{n+1}}}$ in $\boldsymbol{h}$ the waves move at $\boldsymbol{\pm Z}$; no $\boldsymbol{|u|}$, the mesh moves with it
- CFL Criterion, $\boldsymbol{\Delta t^{n+1} = \text{CFL} \min_k (\Delta t_k)}$, $\boldsymbol{\text{CFL} = 0.6}$ a safety factor, as in Method AV (`bitwhiz`: deck key `CFL`, default 0.6)
- Why CFL $\boldsymbol{\leq 1}$, no wave crosses a cell within a step, so neighboring Riemann problems never meet and the frozen $\boldsymbol{Z}$ of Step 2 stays valid; the order 2 half step (Step 1) is stable under the same limit
- Void Cell (`bitwhiz`), $\boldsymbol{c = 0}$: $\boldsymbol{\Delta t_k = \dfrac{\Delta x}{|u_k|}}$, as in Method AV (no limit if it is at rest)
- First and Last Steps, $\boldsymbol{\Delta t^0}$ from the $\boldsymbol{t = 0}$ state ($\boldsymbol{c = c_0}$); the last step is shortened to land exactly on the termination time

Boundary Conditions (Void on Both Ends):

- Ghost Cells, $\boldsymbol{k = 0}$ and $\boldsymbol{k = k_{\max} + 1}$ with $\boldsymbol{P = 0}$, $\boldsymbol{Z = 0}$, $\boldsymbol{u = 0}$ (any finite value; it is multiplied by $\boldsymbol{Z = 0}$)
- Effect, the Step 2 formulas, with nothing changed, give at the left end ($\boldsymbol{Z_L = 0}$; no slope beside a ghost, so $\boldsymbol{u_1^L = u_1}$, $\boldsymbol{P_1^L = P_1}$)

  . $\boldsymbol{u_{\star,\frac{1}{2}} = u_1 - \dfrac{P_1}{Z_1}}$ $\boldsymbol{P_{\star,\frac{1}{2}} = 0}$ the acoustic ***free surface*** (velocity doubling for a shocked state)
- Interfaces, nothing special: each side brings its own $\boldsymbol{Z}$, and each cell calls its own EOS
- Zero Impedance (`bitwhiz`), if $\boldsymbol{Z_L + Z_R = 0}$ (a void cell, $\boldsymbol{c = 0}$, beside a ghost or another void cell): $\boldsymbol{P_\star = 0}$ and $\boldsymbol{u_\star = \frac{1}{2}(u_L + u_R)}$, a ghost taking its neighbor's face $\boldsymbol{u}$ not reached by the class materials

One Cycle, In Order:

a. Face states $\boldsymbol{u^{L,R}}$, $\boldsymbol{P^{L,R}}$ in every cell (order 1: the cell values; order 2: slopes and half step)

b. Riemann solution $\boldsymbol{u_\star}$, $\boldsymbol{P_\star}$ on every face, with $\boldsymbol{Z^n = \rho^n c^n}$ from its two cells

c. Positions $\boldsymbol{x^{n+1}}$, then densities $\boldsymbol{\rho^{n+1}}$

d. Velocities $\boldsymbol{u^{n+1}}$ and total energies $\boldsymbol{E^{n+1}}$, then $\boldsymbol{e^{n+1}}$

e. Pressures $\boldsymbol{P^{n+1}}$ (keep them $\geq \mathbf{0}$; `bitwhiz`: the spall rule), then sound speeds $\boldsymbol{c^{n+1}}$

f. The next time step $\boldsymbol{\Delta t^{n+1}}$

g. Write output if needed; $\boldsymbol{n \leftarrow n+1}$; repeat until the termination time

- Compared with Method AV, no $\boldsymbol{q}$, no node mass, no half-step velocity, no kinetic-energy correction: the Riemann solution at the face replaces all four

bitwhiz: Where This Lives in the Code:

- Deck, `method = RS` and, optionally, `order = 1` or `2` (default 2) in the Solver block; every output names the scheme, e.g. "RS, order 2 (MUSCL-Hancock, Van Leer)"

- Steps, module `rs`: Step 1 → `rs_face_states`; Step 2 → `riemann_face`; Steps 3–4 → `rs_cell_step` ($\boldsymbol{\rho}$, $\boldsymbol{u}$, $\boldsymbol{E}$, $\boldsymbol{e}$) and `rs_state` (EOS, spall rule and $\boldsymbol{c}$, through `cell_pressure`, shared with Method AV); Time Step → `rs_cell_dt`

- One Cycle, `maincode`, contained routine `rs_region`: loop 0, cells (a); loop 1, faces (b, and $\boldsymbol{x^{n+1}}$); loop 2, cells ($\boldsymbol{\rho^{n+1}}$, then d–f); one OpenMP parallel region, as for AV, and the same answer on any number of threads

- Database Slots, faces: $\boldsymbol{x}$ `jx`, $\boldsymbol{u_\star}$ `ju_star`, $\boldsymbol{P_\star}$ `jp_star`; cells: $\boldsymbol{M_k}$ `jm`, $\boldsymbol{\rho}$ `jrho`, $\boldsymbol{u}$ `juc`, $\boldsymbol{e}$ `je`, $\boldsymbol{P}$ `jp`, $\boldsymbol{c}$ `jc`, $\boldsymbol{\mathcal{D}}$ `jdam`, $\boldsymbol{u^L, u^R, P^L, P^R}$ `ju_lf`, `ju_rf`, `jp_lf`, `jp_rf`, and the material constants; $\boldsymbol{Z = \rho c}$ is formed on the spot, not stored

- Outputs, the tracer column `KMPS` is $\boldsymbol{u_\star}$; `Q` is NaN (there is no $\boldsymbol{q}$); the PDV tracer on an interface face reads $\boldsymbol{u_\star}$ there